

Name: NEPTUN: Score: Grade:

Fundamentals of theory of computation I. – exam 2025.05.30.

0-23: grade 1, 24-31: grade 2, 32-39: grade 3, 40-47: grade 4, 48-60: grade 5

1. True or false? (Put an X in the column of the correct answer.) (20x1 points)

true false

- ☐ ☐ For every homomorphism $h: V_1^* \rightarrow V_2^*$ we have $h(\varepsilon) = \varepsilon$.
- ☐ ☐ For every language L we have $L^+ \subset L^*$. ($\subset$: strict inclusion.)
- ☐ ☐ The empty language ($\emptyset$) is a regular language.
- ☐ ☐ All languages that can be recognized by a deterministic finite automaton can be generated by a right linear grammar.
- ☐ ☐ Regular expressions $bc(bc)^*$ and $(bc)^*$ describe the same language.
- ☐ ☐ For every nondeterministic finite automaton A there exists a deterministic pushdown automaton A' with the property $L(A') = L(A)$.
- ☐ ☐ $G = (\{S, A\}, \{c, d\}, \{S \rightarrow AcA, S \rightarrow d, A \rightarrow cdA, A \rightarrow cc\}, S)$ is a grammar of type 3.
- ☐ ☐ Let $S \rightarrow \varepsilon$ and $S \rightarrow SAB$ be two rules of a grammar $G = (N, \Sigma, P, S)$, where $A, B \in N$. Then G can NOT be a context free grammar.
- ☐ ☐ Let $G = (N, \Sigma, P, S)$ be a grammar. If all rules in P is of the scheme either $A \rightarrow vBw$ or $A \rightarrow w$ where $A, B \in N, v, w \in \Sigma^*$ then G is a linear grammar.
- ☐ ☐ Intersection of a regular and a context free language is context free.
- ☐ ☐ $\mathcal{L}_2$ is closed under the operation intersection.
- ☐ ☐ If the initial symbol of a context free grammar G is active then $L(G) \neq \emptyset$.
- ☐ ☐ All context free grammars are context sensitive grammars as well.
- ☐ ☐ If states q and r are indistinguishable ($q \sim r$) in a deterministic finite automaton $A = (Q, \Sigma, \delta, q_0, F)$, all states are reachable from q_0 and $r \in F$ holds then $q \in F$.
- ☐ ☐ Let $G = (N, \Sigma, P, S)$ be a CF grammar and $u = t_1 \cdots t_n \in \Sigma^+$ ($n \geq 1$) be a word. To decide $u \in L(G)$ CYK algorithm computes some sets H_{ij} ($1 \leq i \leq j \leq n$). Then $u \in L(G) \iff H_{1n} \neq \emptyset$.
- ☐ ☐ Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a pushdown automaton. Then $L(A)$ can be generated by a type 3 grammar.
- ☐ ☐ Let $A = (Z, Q, \Sigma, \delta, z_0, q_0, F)$ be a deterministic pushdown automaton. Then $|\delta(z, q, a)| = 1$ holds for every $z \in Z, q \in Q, a \in \Sigma \cup \{\varepsilon\}$.
- ☐ ☐ Let A be a pushdown automaton. Then there exists a deterministic pushdown automaton A' with the property $L(A') = L(A)$.
- ☐ ☐ In derivation trees of words of context free grammars all inner nodes are labeled by a nonterminal.
- ☐ ☐ Let $L = \{b, bb, bba, abab\}$. Then $|L_{bb}| = 1$.

2. Answer the following questions. No explanation is required.

(a) Give the 2nd power of the language described by regular expression b^*a by FORMULA.

(b) List the names of the regular operations.

(c) Let $G_1 = (\{A\}, \{a, b\}, \{A \rightarrow bbA \mid ab \mid \varepsilon\}, A)$ and $G_2 = (\{B\}, \{a, b\}, \{B \rightarrow aB \mid ba\}, B)$ be regular grammars. Give a regular grammar generating $L(G_1)L(G_2)$ according to the construction we have learnt. (Don't forget to name the initial symbol.)

(d) Let $G = (N, \Sigma, P, S)$ be a grammar of type 1. What condition should hold if $S \rightarrow \varepsilon$ is one of the rules of P ?

(e) Let $G = (N, \Sigma, P, S)$ be a grammar és $u, v \in (N \cup \Sigma)^*$. Define what does it mean that v can be derived from u . ($u \Rightarrow_G^* v$ relation)

(f) Let $A = (Q, \Sigma, \delta, Q_0, F)$ be a nondeterministic finite automaton. Define $L(A)$, the language recognized by A .

(g) Consider a nondeterministic finite automaton $A = (\{p, q, r\}, \Sigma, \delta, \{p, r\}, \{q\})$ (i.e., $F = \{q\}$). List all the accepting states of the deterministic finite automaton A' we have learnt to be equivalent with A . (The set of states of A' is the powerset of the set of states of A .)

(h) Let L be a language of type 3. What is the initial state of the Myhill-Nerode automaton A_L^{MN} ? (The automaton with the states corresponding to the suffix languages wrt. the words over the alphabet of L .)

(i) Assume that $H(A) = \{A, B, D\}$ holds in the chain elimination algorithm of CF grammars (Chomsky NF). Let the rules of G be as follows (S, A, B, C, D are nonterminals, a and b are terminals):

$$S \rightarrow AB \mid CD \quad A \rightarrow CC \mid B \quad B \rightarrow A \mid AA \mid D \mid a \quad C \rightarrow A \mid AS \quad D \rightarrow B \mid b$$

Give all the rules of a chain free grammar G' equivalent with G and having A on the left hand side.

(j) Give a pushdown automaton rule with the following effect. The top element of the stack is a b . By an ε -move the automaton changes its state from r to q . The content of the stack remains unchanged.

3.

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(4+5+5+6 points)

- (a) Let L be a language. Define n th power ($n \in \mathbb{N}$), iterative closure, positive closure of L .
- (b) Give the definition of regular expressions and the languages they describe.
- (c) Let $G = (N, \Sigma, P, S)$ be a context free grammar. Introduce the type 2 length reduction algorithm on G . (Don't forget to mention what can be an input and what is the output of the algorithm.)
- (d) Define pushdown automaton (including its components and transition function).